PALTURO · ENGINEERING DOCUMENTATION

Dynamic UX, QOL & Social Presence

Complete record of the notification, feedback, chat quality-of-life, privacy, presence, typing, and read-receipt work.

Document	Full change documentation
Branch	ui-features
Baseline	`3ab6700`
Final commit	`08de0e8`
Date	5 November 2026
Scope	54 commits · 94 files · 155 tests
Status	Tracks A, B, C complete; D-01 complete; A5 blocked on credentials; D-02 deferred

Prepared from the repository history, migrations, persistent plan, audit sessions, and final verification runs. Secrets and credentials are intentionally excluded.

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> **Reading note.** Phase IDs match the persistent plan (A0-A7, B0-B3, C0-C3, D-01/D-02). A phase marked VERIFIED passed three independent subagent audits: functionality, quality, and regression.

1. Executive summary

This release turns PalTuro's notification layer into a complete dynamic UX system, then adds the day-to-day chat and list controls expected from a modern messaging product, and finally introduces privacy-controlled presence, typing, and read receipts.

Commits	54 (from `3ab6700` to `08de0e8`)
Files changed	94 (94 files changed, 9635 insertions(+), 1301 deletions(-))
Test suite	155 tests passing
Builds	Web release and Android debug APK pass
Database	7 forward-only migrations, applied to the linked Supabase project
Audits	14 gated phases, each with 3 PASS audits (A0-A4, A6, B0-B3, C0-C3)

What users can now do

- See live request, match, feedback, moderation, and message events as non-blocking top banners, with a subtle match moment and an Activity inbox.
- Control notifications by category, previews, quiet hours, sound, and vibration, and see honest permission state.
- Receive local system notifications while the app is alive, including while backgrounded on Android, with tap and cold-start routing.
- Refresh lists, work offline with a clear strip, see an app badge, and keep unsent drafts per chat.
- Mute a chat, mark it unread, copy a message, search the thread, jump to the latest message, retry failed sends, and see the first unread divider.
- Manage chats from a long-press sheet and triage Activity with filters, day groups, swipe-to-read, and clear-read.
- Control online status, typing, and read receipts for Pals only; see Online/last-seen, typing dots, and Seen only when privacy allows.

> **Not shipped.** Push delivery to killed or suspended apps (A5) remains blocked on Firebase/APNs/VAPID credentials. The app does not claim killed-app delivery or guaranteed iOS background delivery. See section 9.

2. Scope and tracks

Track	Theme	Phases and outcome	Status
A	Dynamic UX core	A0 notification foundation, A1 banners and match moment, A2 haptics/sound, A3 preferences, A4 local notifications, A5 push (blocked), A6 feedback audit, A7 release verification	A0-A4, A6, A7 done A5 blocked
B	Quality of life	B0 refresh/offline/badge/drafts, B1 chat QOL, B2 list and Activity QOL, B3 contextual polish	Done
C	Realtime social	C0 privacy contracts, C1 presence, C2 typing, C3 read receipts	Done
D	Optional polish	D-01 what's new, D-02 review prompt	D-01 done D-02 deferred

Work was gated phase by phase. No phase advanced without functionality, quality, and regression audits returning PASS in separate subagent sessions.

3. Track A - Notification experience

A0 - Notification Foundation

Commits: `8e0f5f3, 5aed01e` · Status: **VERIFIED**

Goal. Introduce a typed, realtime notification backbone so every later notification surface uses one source of truth.

- Added AppNotification with known kinds (request, accepted, declined, feedback, moderation, unknown), defensive parsing, payload fallbacks, read state, and copyWith/toRow adapters.
- Added NotificationService: notifications realtime channel, dedupe by id, unread count, mark-one/mark-all, lifecycle refresh, sign-out reset, and reconnect handling.
- Added AppRouter and a global navigator key so banners, Activity, local notifications, and cold starts can open Activity, Requests, Chat, or Profile before or after the first frame.
- Replaced ActivityService with NotificationService and migrated every caller.
- Server: notifications added to the supabase_realtime publication with a SELECT grant scoped by the existing notif_read policy; list_notifications enriched with actor name/avatar, skill, role, and connection status.

Key files: `lib/models/app_notification.dart`, `lib/services/notification_service.dart`, `lib/services/app_navigation.dart`, `lib/services/app_navigator.dart`, `supabase/migrations/20261105000100_notifications_realtime.sql`, `supabase/migrations/20261105000200_notification_context.sql`

Verification: 91 tests; analyzer clean; live RPC and grant probes; three subagent audits PASS at 5aed01e.

A1 - In-app Dynamic UX

Commits: `45cf7e7, be90c0d, de4046e, c480fa7, 06f10d8` · Status: **VERIFIED**

Goal. Make live events visible without leaving the current screen, with a subtle top banner and a celebratory match moment.

- Added a global AppBanner queue: one banner at a time, 4s default, close/swipe dismiss, tap routing, live-region semantics, reduced-motion support, and grouping per conversation.
- Added the match banner: 'You're now Pals with X', skill line, Say hello, about 7s, once per connection, persisted across restarts; it pre-empts a standard banner and restores the queued one.
- Replaced the old bottom match sheet with the top banner.
- Emitted IncomingMessageEvent from ChatStateService only for messages outside the active room and not authored by the current user; same-conversation messages coalesce.
- Added the Activity bell on For You with unread count, and route-aware suppression for Activity, pushed Requests, and the active chat.
- Hardened realtime auth: refresh expired sessions before joining, resubscribe with backoff on JWT/channel errors, track joined state, and recover on token refresh or resume.

Key files: `lib/widgets/app_banner.dart`, `lib/widgets/match_banner.dart`, `lib/services/notification_presenter.dart`, `lib/services/app_shell_state.dart`, `lib/models/incoming_message_event.dart`, `lib/services/chat_state_service.dart`, `lib/screen/home_page.dart`, `lib/screen/request_page.dart`

Verification: 99 tests; live light/dark banner, match, and message probes; three subagent audits PASS at 06f10d8.

A2 - Haptics and Sound

Commits: `b96c571, 4de648a` · Status: **VERIFIED**

Goal. Give interactions meaningful physical and audio feedback without double-cueing or ignoring user preferences.

- Rewrote Haptics as named intents: selection, tap, toggle, impact, success, warning, error, match; 50ms per-intent throttle; web/desktop no-op; Vibration preference gate.
- Added SoundService reusing existing assets: messages.wav, newmatch.wav, splash.wav; sound preference gate, lifecycle gate, 150ms debounce, serialized init, stop on route change.
- Wired feedback for matches, messages, requests, declines, sends, undo, profile save, settings toggles, filters, and destructive confirmations.
- Removed raw HapticFeedback calls from request cards so the preference cannot be bypassed; removed the duplicate decline cue.

- Settings enabling Sound or Vibration previews the choice.

Key files: `lib/services/haptics.dart`, `lib/services/sound_service.dart`, `lib/services/notification_presenter.dart`, `lib/main.dart`, `lib/onboarding/onboarding_complete.dart`

Verification: 106 tests; 4 haptic + 3 sound tests; single-cue audits; three subagent audits PASS at 4de648a.

A3 - Notification Preferences

Commits: `75b1ff8, 72fe744, f89f540` · Status: **VERIFIED**

Goal. Give users honest, per-category control over alerts, previews, quiet hours, sound, and vibration.

- Expanded PreferencesService into an observable model: master switch, messages, requests/matches, activity, message previews, quiet hours window, sound, vibration.
- Presenter gates each event kind, suppresses sound/haptics during quiet hours while still showing banners, and replaces message text with 'New message' when previews are off.
- Added a dedicated Notifications settings page: delivery, permission status, categories, preview example, sound/vibration with previews, adaptive quiet-hours time pickers, and Pals-only context.
- Master off disables categories without erasing their values; Settings root shows an accurate summary.
- Removed fictional OS-permission copy and scoped quiet hours to notification feedback only.

Key files: `lib/services/preferences_service.dart`, `lib/screen/settings/notification_preferences_page.dart`, `lib/screen/settings.dart`, `lib/services/notification_presenter.dart`

Verification: 114 tests; live light/dark settings screenshots; 360/200% coverage; three subagent audits PASS at f89f540.

A4 - Local Notifications

Commits: `dc5c03c, a2c3fc6, ed9c0ed` · Status: **VERIFIED**

Goal. Deliver system notifications while the process is alive, with permissions, channels, and tap routing.

- Added flutter_local_notifications 22.3.3, Android core-library desugaring, POST_NOTIFICATIONS and VIBRATE, and ic_stat_palturo status icon.
- Added LocalNotificationService: initialization, four Android channel combinations per category (sound/vibration independently), web/iOS setup, foreground suppression, category/quiet/preview gating, logout cancel.
- Permission status model distinguishes not-asked from denied; request from Settings; denied opens system settings; state refreshes on resume.
- Tap and cold-start payloads route to Chat, Requests, Activity, or Profile with peer context.
- iOS: UNUserNotificationCenter delegate set in AppDelegate; deployment target raised to 13.0.

Key files: `lib/services/local_notification_service.dart`, `android/app/build.gradle.kts`, `android/app/src/main/AndroidManifest.xml`, `android/app/src/main/res/drawable/ic_stat_palturo.xml`, `ios/Runner/AppDelegate.swift`, `ios/Runner.xcodeproj/project.pbxproj`

Verification: 120 tests; web release + Android debug APK; live permission screenshot; three subagent audits PASS at ed9c0ed.

A5 - Push Delivery

Commits: `none` · Status: **BLOCKED**

Goal. Deliver alerts to killed or suspended apps through FCM and APNs.

- Not implemented. No Firebase project, google-services.json, GoogleService-Info.plist, APNs .p8/Key ID/Team ID, web VAPID key, or FCM service account is available.
- The app must not claim killed-app delivery or guaranteed iOS background delivery.
- Everything except killed-app push is covered by A4 local notifications plus server reconciliation on resume.

Key files: `Would require: Firebase project (Android/iOS/web)`, `android/app/google-services.json`, `ios/Runner/GoogleService-Info.plist`, `APNs auth key + Key ID + Team ID`, `Web VAPID public key`, `Firebase service account for the edge function`

Verification: BLOCKED on external credentials. See Accepted delivery gap in section 9.

A6 - Feedback System Audit

Commits: `8d8f9e6, 445c4c9, 5b97786, da7ff16` · Status: **VERIFIED**

Goal. Unify user feedback and remove technical error leakage.

- Added AppFeedback: one snackbar contract with info/success/error tone, icon, durations (4s/3s/6s), one action, clearSnackBars, and a ScaffoldMessenger variant for dialogs.
- Adopted the helper across home, chat, matches, requests, block list, explore, account, and settings feedback paths.
- Friendly error mapping now rejects SQL/driver text (Postgrest, SQLSTATE, varying(), auth) before the allowlist, and preserves validation copy including password digit rules and account-deletion messages.
- Undo snackbars use an explicit 5s duration; account password/delete fallbacks are distinct and accurate.

Key files: `lib/widgets/app_feedback.dart`, `lib/utils/friendly_error.dart`, `lib/screen/home_page.dart`, `lib/screen/chats_room.dart`, `lib/screen/request_page.dart`

Verification: 122 tests; negative technical-leak tests; three subagent audits PASS at 5b97786/da7ff16.

A7 - Release 1 Verification

Commits: `verification only` · Status: **VERIFIED**

Goal. Prove the A track is release-ready.

- Analyzer clean, 122/122 tests, release web build, Android debug APK.
- Visual matrix at 360/390/1440 and dark mode (home, settings, Activity, banner states).
- Live web probes for request, match, message, and permission states.
- Accepted delivery gap documented in the plan before B/C work continued.

Key files: `Verification artifacts in the persistent plan`, `No source changes in this phase`

Verification: All gates green; A5 remains blocked.

4. Track B - Quality of life

B0 - QOL Primitives

Commits: `cb7b96a, 27c3355, 9df1965` · Status: **VERIFIED**

Goal. Make the app feel current under poor connectivity and daily use.

- Added pull-to-refresh to Chats, Activity, Requests (incoming/sent), and Browse; refresh keeps data and scroll position when content exists and shows a snackbar on failure instead of replacing the list.
- Added ConnectivityService and a persistent offline strip laid out above app content (not over the app bar), with reconnect refresh for notifications and conversations.
- Added AppBadgeService: launcher badge = activity unread + chat unread, gated by the master preference, skipped on web/unsupported, serialized writes, reset on sign-out.
- Added DraftService: per-conversation local drafts, 350ms debounce, restore on open, clear on send/sign-out, 'Draft:' preview in the chat list.

Key files: `lib/services/connectivity_service.dart`, `lib/services/app_badge_service.dart`, `lib/services/draft_service.dart`, `lib/widgets/offline_banner.dart`, `lib/screen/chats_page.dart`, `lib/screen/chats_room.dart`

Verification: 131 tests; web + APK; live offline strip and draft screenshots; three subagent audits PASS at 9df1965.

B1 - Chat QOL

Commits: `a24048f, 4b18f8a, fdd641b` · Status: **VERIFIED**

Goal. Make everyday chat management fast and private.

- Per-chat mute with 1h/8h/1d/1w/always and unmute; muted chats suppress banners and local notifications but still count unread; mute icon in the list; server-persisted.
- Mark unread: server flag, red dot in the list, clears when the chat opens, does not affect push.
- Draft integration in the composer and list preview.
- Scroll-to-bottom FAB with new-message count; auto-scroll only when already at the bottom.
- Long-press copy with clipboard + one snackbar; in-chat search with results, highlight, and jump; unread divider at the first unread message.
- Retry-all failed pending messages sequentially, stop on renewed failure; reconnect triggers retry.

Key files: `lib/services/chat_state_service.dart`, `lib/screen/chats_room.dart`, `lib/screen/chats_page.dart`, `lib/widgets/chat_search_sheet.dart`, `lib/widgets/chat_menu_row.dart`, `supabase/migrations/20261105000300_connection_preferences.sql`

Verification: 138 tests; live mute/mark-unread/draft/divider/FAB/copy/search probes; three subagent audits PASS at fdd641b.

B2 - List and Activity QOL

Commits: `a25dacd, b8419ca, af0f3c4` · Status: **VERIFIED**

Goal. Bring chat management and Activity triage to the list level.

- Chat long-press sheet: View profile, Mute/Unmute, Mark unread, Report, Block; profile dialog keeps block/report paths.
- Activity filters All/Unread/Requests/Updates with counts, day groups (Today, Yesterday, date), filtered empty state, and clear-search recovery.
- Swipe an unread row to mark it read; Clear read confirms and calls a server RPC that deletes only the caller's read rows (unread rows stay).
- Failure signalling for clear/mute/unread now reports errors instead of false success.
- Saved filters per screen: chat role/status and request incoming/sent filters persist, restore on open, and clear actions reset them.

Key files: `lib/screen/notifications_page.dart`, `lib/screen/chats_page.dart`, `lib/widgets/chat_mute_sheet.dart`, `lib/services/preferences_service.dart`, `supabase/migrations/20261105000400_clear_read_notifications.sql`

Verification: 141 tests; live Activity filters/groups/clear confirm and chat long-press; three subagent audits PASS at af0f3c4.

B3 - Contextual Polish

Commits: `2491fc3, 8110b51, 3e11df1, 12afa6e, 26d38a0, 708316e, bb54994` · Status: **VERIFIED**

Goal. Surface the right next step without nagging or dead ends.

- Continue-setup card appears only when setup is incomplete and onboarding was skipped or a draft exists; Resume restores the review step; dismiss persists per user; it reappears after skipping again.
- Profile completion nudge after a successful request uses the app banner (non-blocking, SafeArea, queued, closable, 10s) and shows at most once per session; the request is already committed.
- Empty states now each have one clear action: start swiping, view sent requests/refresh, view requests/show all, update skills/refresh categories.
- Profile shows only one completion prompt at a time; chats filtered empty has a clear action; Requests never traps the tab in sent view.

Key files: `lib/widgets/resume_setup_card.dart`, `lib/screen/profile_page.dart`, `lib/screen/home_page.dart`, `lib/screen/chats_page.dart`, `lib/screen/request_page.dart`, `lib/screen/explore_page.dart`

Verification: 145 tests; 360/200% card coverage; live nudge banner and dismissal; three subagent audits PASS at bb54994.

5. Track C - Privacy and realtime social

C0 - Privacy Contracts

Commits: `a79ed63`, `5c1a4fd`, `4aaaa02`, `4021a43`, `3fa4a45` · Status: **VERIFIED**

Goal. Define what status data exists and who may see it before adding presence/typing/receipts.

- Profiles gained last_seen_at, presence_enabled, typing_enabled, read_receipts_enabled; no direct client column grants.
- Added touch_last_seen (self-only, only when presence enabled), set/get_chat_privacy, and get_pals_presence (accepted Pals only, last_seen masked when off).
- get_conversations exposes peer presence/typing/receipts/read fields, masked to accepted connections; read timestamps are hidden when receipts are off.
- Revoked table SELECT on connections and granted only non-receipt columns, closing the direct-read bypass for read timestamps.
- Chat privacy page with three switches and 'Pals only'; switches are disabled until server state loads, load failures show Retry, loads coalesce, and reset is generation-safe.

Key files: `supabase/migrations/20261105000500_chat_privacy.sql`, `supabase/migrations/20261105000600_privacy_status_masking.sql`, `supabase/migrations/20261105000700_lock_connection_reads.sql`, `lib/services/chat_privacy_service.dart`, `lib/screen/settings/chat_privacy_page.dart`

Verification: 149 tests; live ACL and masking probes; 360/200% page test; three subagent audits PASS at 3fa4a45.

C1 - Presence

Commits: `69bec41`, `c8bb845`, `38949d7`, `b876970`, `487b85a`, `5829b91`, `27f4784` · Status: **VERIFIED**

Goal. Show whether a Pal is online without leaking status or keeping stale state.

- 60s foreground heartbeat + conversation refresh while resumed; heartbeat is disabled when privacy is unloaded or presence is off; stops on pause/sign-out.
- Chat list shows a green Online or muted Active Nm/h/d only when the peer allows presence; message time remains visible.
- Chat room subscribes to chat:<connectionId> presence, tracks only with presence enabled, filters the peer by exact user id, and removes channels on leave/privacy-off/error without re-entrant teardown.
- Privacy toggle off stops broadcast and hides status; re-enabling retracks; typing-only mode still joins the channel.

Key files: `lib/services/chat_state_service.dart`, `lib/screen/chats_room.dart`, `lib/screen/chats_page.dart`, `lib/utils/presence_label.dart`

Verification: 150 tests; live Online in list and room; three subagent audits PASS at 5829b91.

C2 - Typing Indicators

Commits: `354ff68`, `8da9b57`, `5bb26dc` · Status: **VERIFIED**

Goal. Show that a peer is composing without database writes or leaking to non-members.

- Typing broadcasts on the existing chat channel; about 2s throttle, 3s peer expiry, stop on empty/send/leave; no DB writes.
- Typing indicator under the peer name: three animated dots, static fallback when animations are disabled, live-region label.
- Gated by the local typing setting and the peer's typing flag; works with presence off (channel joins, no presence track).
- Typing-only channels recover after errors; typing appears before the first message; the pre-first-message match card stays while typing.

Key files: `lib/widgets/typing_indicator.dart`, `lib/screen/chats_room.dart`

Verification: 152 tests; typing payload and 200%/reduced-motion tests; three subagent audits PASS at 5bb26dc.

C3 - Read Receipts

Commits: `879e593`, `3f468e5`, `98e3b59` · Status: **VERIFIED**

Goal. Show Seen only when the peer has actually read and allows receipts.

- Conversation contract carries other_read_at and other_read_receipts_enabled; unread counts unchanged; dead is_read handling removed.
- Seen renders only under the last own message, only when the peer's read timestamp covers it and the peer allows receipts.
- Live receipts use a throttled read broadcast on the chat channel (trailing flush), then reload the server timestamp; displayed time is server-authoritative, avoiding device-clock skew.
- Read broadcasts are paused while backgrounded, peer-scoped on receive, and do not echo; 60s refresh reconciles offline peers.

Key files: `lib/screen/chats_room.dart`, `lib/services/chat_state_service.dart`

Verification: 154 tests; live Seen screenshot; three subagent audits PASS at 98e3b59.

6. Track D - Optional polish

ID	Feature	Implementation	Status
D-01	What's new	Version-gated sheet with three bullets, dismiss persists per version, scrollable at 360/200%. Commit `08de0e8`.	Done
D-02	Review prompt	Not implemented. A review prompt without a published store listing would be a fake action or dead link. Gate design remains: positive exchange, once per version, never over a dialog.	Deferred

7. Architecture and new modules

The work stays within the existing Flutter app and Supabase backend. Notifications have one typed source, one presenter, and one local delivery service; chat privacy/status lives on the profile and is exposed through RPCs; realtime chat topics are reused for presence, typing, and read events.

Services

Module	Responsibility
`NotificationService`	Typed notification state, realtime channel, dedupe, unread list/count, mark read, lifecycle, sign-out reset.
`NotificationPresenter`	Chooses banner kind, applies category/quiet/preview/privacy gates, plays match/message feedback, suppresses the active screen.
`LocalNotificationService`	System notifications, Android channels (four sound/vibration combinations per category), permission state/request, tap and cold-start routing.
`AppBadgeService`	Launcher badge from activity + chat unread, preference gate, serialized writes, web no-op, sign-out reset.
`ConnectivityService`	Online state, offline strip source, reconnect refresh for notifications and conversations.
`DraftService`	Per-conversation drafts, debounce, restore, clear on send/sign-out.
`ChatPrivacyService`	Server-backed presence/typing/receipt switches, coalesced load, optimistic save with revert, reset on sign-out.
`Haptics`	Named intents, throttle, platform and preference gates, match pattern.
`SoundService`	Shared player over messages/newmatch/splash assets, preference/lifecycle gates, debounce, serialized init.
`AppShellState`	Selected tab, Activity/Requests open flags for route-aware suppression.
`AppRouter / navigatorKey`	Queue-safe routing from services and cold starts.

Primary screens

Screen	Change
Activity	Typed list, filters All/Unread/Requests/Updates with counts, day groups, swipe-to-read, clear-read, realtime updates.
Chats	Pull-to-refresh, presence, mute icon, manual unread dot, draft preview, long-press actions, clear search/filters.
Chat room	Banners/suppression, draft composer, scroll FAB with count, copy, search, unread divider, mute/mark unread, presence, typing, Seen.
Notifications settings	Delivery, permission, categories, previews, sound/vibration previews, quiet hours, master gating.
Chat privacy	Online status, typing, read receipts, Pals-only, load/error gating.
Profile	One completion prompt at a time, resume setup, scoped dismissal.
Requests	Pull-to-refresh, saved filters, clear actions, no tab dead ends, backable sent view.
Browse	Pull-to-refresh, mentor/learner role, empty refresh action.

Event flow

- **Server notification** → notifications realtime insert → NotificationService (typed, deduped) → NotificationPresenter → top banner; Activity list and unread count update.
- **Foreground message** → ChatStateService realtime → IncomingMessageEvent (outside the active room) → presenter → banner with sender + preview; local system notification only when not foreground.

- **Background/alive local notification** → LocalNotificationService → Android channel chosen by category and sound/vibration → system notification → tap payload → AppRouter.
- **Chat topic** → presence state (Online), typing broadcast (2s throttle/3s expiry), read broadcast (throttled, server-authoritative reload).

8. Database and server changes

Seven migrations were added and applied to the linked Supabase project. All functions are SECURITY DEFINER with a pinned `search_path`; writes are RPC-only; the notification and connection tables remain SELECT-scoped by RLS.

#	Migration	Purpose	Objects and behaviour
001	`20261105000100_notifications_realtime.sql`	Notifications realtime	Adds public.notifications to supabase_realtime (guarded) and grants SELECT to authenticated; row visibility stays scoped by notif_read (user_id = auth.uid()).
002	`20261105000200_notification_context.sql`	Activity context	Recreates list_notifications with actor name/avatar, skill, role, and connection status joins while keeping payload as the source of ids.
003	`20261105000300_connection_preferences.sql`	Chat mute + manual unread	Adds connection_preferences (user_id, connection_id, muted_until, marked_unread), SELECT-only RLS, set_connection_mute, set_connection_unread, get_connection_preferences, and appends muted/marked fields to get_conversations.
004	`20261105000400_clear_read_notifications.sql`	Clear read Activity	Adds clear_read_notifications(), a SECURITY DEFINER delete limited to user_id = auth.uid() and read_at is not null.
005	`20261105000500_chat_privacy.sql`	Privacy contracts	Adds last_seen_at and three privacy flags to profiles; adds touch_last_seen, set/get_chat_privacy, get_pals_presence; appends peer presence/typing/receipts/read fields to get_conversations. Realtime topic ACL cannot be installed from the management role (see limitations).
006	`20261105000600_privacy_status_masking.sql`	Status masking	Recreates get_conversations so presence/typing/receipt flags and last_seen/read timestamps are returned only for accepted connections, and read_at is null when receipts are off.
007	`20261105000700_lock_connection_reads.sql`	RPC-only read timestamps	Revokes table SELECT on connections from anon/authenticated and grants only id, requester_id, recipient_id, status, created_at, responded_at, skill_name, skill_path; read timestamps are reachable only through RPCs.

Security model

- **Notifications:** authenticated can SELECT their own rows (notif_read); all writes go through SECURITY DEFINER RPCs (mark read, clear read).
- **Connections:** authenticated can read only non-receipt columns; read timestamps are RPC-only. RLS limits rows to participants.
- **Connection preferences:** SELECT-only for the owner; mute/unread writes go through validated RPCs.
- **Profiles:** presence/typing/receipt columns have no direct client grants; status flows through touch_last_seen, set/get_chat_privacy, get_pals_presence, and get_conversations.
- **Status masking:** only accepted connections receive presence/typing/receipt flags and timestamps; last_seen is null when presence is off; peer read_at is null when receipts are off.

> **Read receipts transport.** Publishing `connections` would require table-wide SELECT for Realtime and would re-expose the read timestamps that C0 locked down. C3 therefore uses a read broadcast on the chat topic plus the 60s conversation refresh, which is server-authoritative and reconciles offline peers.

9. Delivery and privacy behaviour

Notification delivery matrix

Situation	Delivery
App open, any tab	In-app banner, badge, haptics/sound; Activity and unread update.
App open, target screen	Badge/Activity only; no duplicate banner for the screen in view.
App backgrounded, process alive on Android	Local system notification (best effort), with tap routing.
App backgrounded on iOS	Not guaranteed; the OS may suspend the app/socket.
App killed or force-closed	No notification until the app is reopened.
App reopened after missed events	Activity, unread counts, and messages reconcile from the server.
Broadcast announcement to inactive users	Not delivered until the next app open.

> **Accepted gap (A5).** Push to killed/suspended apps requires FCM/APNs/VAPID credentials. Do not claim killed-app alerts or guaranteed iOS background delivery in release notes. There is no data loss: notification rows, messages, and unread counts are server-backed.

Mute and unread

- Mute durations: 1 hour, 8 hours, 1 day, 1 week, until turned back on; Unmute appears when muted.
- Muted chats produce no banner or local notification; unread count and launcher badge still count them.
- Mark unread is a server flag; the dot clears when the chat opens; opening also advances the server read cursor used for Seen.

Typing and presence privacy matrix

Show me online	Typing	Receipts	Typing channel	Presence track	Read broadcast
On	On	On	Yes	Yes	Yes
On	Off	On	Yes	Yes	Yes
Off	On	On	Yes	No (untracked)	Yes
Off	Off	Off	No channel	No	No

Receiving peer typing/read status is additionally gated by the peer's own flags returned from get_conversations.

Seen rules

- Only the last own message can show Seen.
- Seen requires the peer's read timestamp to be at or after that message's server created_at.
- If the peer disabled read receipts, no timestamp is exposed and Seen is hidden.
- Live read events trigger a server reload; the payload clock is never displayed.

10. Platform configuration

Dependencies added

Package	Version	Why
`flutter_local_notifications`	^22.3.3	Local system notifications, channels, permission APIs, tap payloads.
`connectivity_plus`	^7.3.2	Offline strip and reconnect refresh.
`app_badge_plus`	^1.3.5	Launcher badge for combined unread.

Android

- `isCoreLibraryDesugaringEnabled = true` and `desugar_jdk_libs:2.1.4` for the notification plugin.
- `POST_NOTIFICATIONS` and `VIBRATE` permissions in the manifest.
- Status icon `ic_stat_palturo`.
- Channels: `palturo_messages`, `palturo_requests`, `palturo_activity`, plus `_sound`, `_vibration`, and `_quiet` variants per category.

iOS

- `UNUserNotificationCenter.current().delegate = self` in AppDelegate so foreground and cold-start responses reach the plugin.
- Deployment target raised from 12.0 to 13.0 (framework plist and all project configurations).

Web

- Firebase messaging is not configured (A5 blocked). Local notification initialization and permission APIs are wired; unsupported browsers fail closed.
- Launcher badges are skipped on web.

11. Verification and audits

Analyzer	`flutter analyze --no-pub` - clean
Tests	`flutter test --no-pub` - 155/155 passing
Web	`flutter build web --release --dart-define-from-file=.env` - success
Android	`flutter build apk --debug --dart-define-from-file=.env` - success
iOS	Source/config only on this Windows host; a macOS build is still required.

Test coverage by area

Area	Files	Focus
Notification model/realtime	`app_notification_test.dart`	Parsing, unknown kinds, payload fallbacks, read state, presenter adapter.
Banners	`app_banner_test.dart`	Queue, action, dismiss, match preempt, message coalesce, 360/200%, dark surface.
Haptics/sound	`haptics_test.dart`, `sound_service_test.dart`	Intent mapping, preference gate, throttle, real asset presence, volumes.
Preferences	`notification_preferences_test.dart`	Defaults, persistence, quiet-hours windows, category gating, preview privacy, 360/200%.
Local notifications	`local_notification_payload_test.dart`	Deterministic ids, channel matrix, permission state, routing payloads.
Feedback/errors	`app_feedback_test.dart`, `friendly_error_test.dart`	One-at-a-time snackbars, undo, technical leakage, validation preservation.
QOL services	`qol_services_test.dart`, `draft_service_test.dart`, `offline_banner_test.dart`	Badge math, connectivity transitions, draft reload/clear, offline strip semantics/app-bar taps.
Chat QOL	`chat_qol_test.dart`, `chat_widgets_test.dart`	Mute window, manual unread, divider index, search sheet lifecycle, menu 200%, presence/time row, typing payloads/indicator, Seen gating.
Privacy	`chat_privacy_test.dart`	Presence labels/window, page render 200%, load-failure gate, keys.
Polish	`b3_polish_test.dart`, `whats_new_test.dart`	Resume gate/keys, scoped dismissal, card 360/200%, what's new bullets 360/200%.

Live probes performed

- Notification insert → top banner (light/dark), request accepted → match banner, message insert → message banner.
- Mute sheet → muted list icon; mark unread → list dot → open clears it; draft typed → 'Draft:' preview after back.
- Unread divider visible before unread rows; scroll-up + peer insert → FAB count; long-press copy → clipboard + snackbar; search → highlight/jump.
- Activity filters/counts/day groups; clear-read confirmation; chat long-press action sheet.
- Offline strip above the app bar; reconnect restores content; profile nudge banner; one-time what's new.
- Presence: peer last_seen set → Online in list and room; own last_seen advanced after app load.
- Read receipts: Seen under the last own message.

Three-lens audits

Every implemented phase passed three independent subagent audits. A0-A4/A6, B0-B3, and C0-C3 are recorded below; A5 and D-02 have no audit because they were not implemented.

Phase	Date	Functionality	Quality	Regression	Gate commit

A0	2026-11-04	ses_e592a7f30ffeq8fUwgk2ANDfUM	ses_e592a7f2effeLn8rVWs2rrQPbq	ses_e592a7f2cffe8GfwwpD640Klre	5aed01e
A1	2026-11-04	ses_e5901577affeq2wa4Giny1Zc2G	ses_e59015779ffeOb8RHyB6BUeNZe	ses_e59015777ffeEdT9T7s7tkrkxz	06f10d8
A2	2026-11-04	ses_e58dab2d7ffeux946Z2eIEhyzD	ses_e58dab2d5ffezFyNm7D75VvxET	ses_e58dab2d3ffeV5XsamgW1U0F5L	4de648a
A3	2026-11-05	ses_e58c5095fffegIgQeovEXrHzdE	ses_e58c5095effeUezhvpt8d7r2sM	ses_e58c5095cffegDv70umgaruv8T	f89f540
A4	2026-11-05	ses_e58adc0d1ffeVWtFpai92ukhG8	ses_e58adc0ceffedzJvP1urIo5eUn	ses_e58adc0caffe3Ux8H0s31HmBs0	ed9c0ed
A6	2026-11-05	ses_e588dbc5affeGDZTXb8Pmwvtnw	ses_e588dbc48ffeXisBr5i8JIysPv	ses_e588dbc45ffesfpI7VGMo097lk	5b97786
B0	2026-11-05	ses_e5843f5e2ffefz7EXGxqIi7bLv	ses_e5836b58bffeKweLZoQIitxgHT	ses_e58300b0bffeTYwA6tCsWyiY2M	9df1965
B1	2026-11-05	ses_e58162853ffeY6eXrwW9epCzOH	ses_e58102699ffe7qOQlEUSgH3ZZD	ses_e58000898ffe0vOEXUSh9pzpQy	fdd641b
B2	2026-11-05	ses_e57ef735fffeE9Nt6jlx4U5ATu	ses_e57e2ef40ffeWacaCAK9NOF8US	ses_e57d3680bffeb5fsW9dx9hHZ45	af0f3c4
B3	2026-11-05	ses_e57c785d2ffe2ZSlK7cqlhTQff	ses_e57bc1a33ffefghRV09XmL0825	ses_e57920c5bffeSS8Rni62EB084P	bb54994
C0	2026-11-05	ses_e577e7e0bffeGsh00O8TG4Aap5	ses_e5771c409ffeX81Tj3KRgUxjeZ	ses_e575d14d5ffeer0oF029QXVh3t	3fa4a45
C1	2026-11-05	ses_e56cb7616ffekYx0WUloT2M0zV	ses_e56bd7611ffeNnX2Oe0ucXYyKl	ses_e56a791a8ffesn8nk0lS0zWOFT	5829b91
C2	2026-11-05	ses_e5699eeebffef3BcJw4yGqjhig	ses_e56930aa7ffe3BEpK6pz3eqfd4	ses_e568a4196ffeROc75CIa5nBffl	5bb26dc
C3	2026-11-05	ses_e567e3b64ffe16SQgNbF6cUg0W	ses_e56783fa9ffeX0OJ37fR5RdQMt	ses_e5669814 7ffezN0ys13sCwpNYN	98e3b59

12. Known limitations and deferred work

Item	Status and detail
A5 push	Blocked on Firebase project/config, APNs key, web VAPID, and a service account. Killed-app alerts and guaranteed iOS background delivery are unavailable. In-app, Activity, badges, foreground banners, local notifications while alive, and server reconciliation remain.
Realtime topic ACL	The management role cannot alter realtime.messages (owned by supabase_realtime_admin). chat:<connectionId> topics rely on unguessable UUIDs and client-side privacy gating. Add a dashboard-authored policy if server-side ACLs are required.
Read-receipt transport	C3 uses a read broadcast + 60s conversation refresh instead of publishing connections, because C0 made connection read timestamps RPC-only to honor the receipt toggle.
iOS build	Source/config verified but not compiled on this Windows host. A macOS build and device pass are still required.
D-02 review prompt	Deferred until a store listing exists.
Background receipts	A read event can arrive before mark_messages_read commits; server refresh reconciles within the next poll.

> **Release wording.** The accurate claim is: notifications work in-app and for a background process on Android; push for killed apps is prepared but requires Firebase/APNs credentials.

13. Operations and credential checklist

Build and test

- `flutter pub get`
- `flutter analyze --no-pub`
- `flutter test --no-pub`
- `flutter build web --release --dart-define-from-file=.env`
- `flutter build apk --debug --dart-define-from-file=.env`

Configuration is loaded from `.env` through `--dart-define-from-file`; secrets are not bundled as assets. Migrations live in `supabase/migrations` and are forward-only.

To unblock A5

Credential / asset	Purpose
`Firebase project`	Android, iOS, web apps registered to the release application id.
`android/app/google-services.json`	Android Firebase identity.
`ios/Runner/GoogleService-Info.plist`	iOS Firebase identity.
`APNs .p8 + Key ID + Team ID`	Apple push authorization.
`Web VAPID public key`	Browser push subscription identity.
`Firebase service account JSON`	Edge function authentication to FCM HTTP v1.

The Android application id is currently `com.example.palturo`. Set the production id before creating the Firebase app; changing it later means recreating the Firebase app and push configuration.

Rollback and migration notes

- Migrations are add/replace oriented and re-runnable where practical; connection_preferences and privacy columns use IF NOT EXISTS.
- Function replacements preserve grants; the connections grant change is intentionally restrictive.
- If a migration must be rolled back, restore the prior function body from the previous migration and re-issue grants; do not widen connection read access.

Appendix A - Commit log

Chronological, from baseline 3ab6700 to 08de0e8.

Commit	Date	Phase	Subject
8e0f5f3	2026-11-04	A0	feat(dynamic-ux): add notification event foundation
5aed01e	2026-11-04	A0	fix(dynamic-ux): grant notification reads and refresh activity enrichment
45cf7e7	2026-11-04	A1	feat(dynamic-ux): add in-app banners, match banner, and message events
be90c0d	2026-11-04	A1	test(dynamic-ux): cover banner at 360 and 200% text
de4046e	2026-11-04	A1	fix(dynamic-ux): wire banner actions and isolate live state
c480fa7	2026-11-04	A1	fix(dynamic-ux): recover realtime channels after failed joins
06f10d8	2026-11-04	A1	fix(dynamic-ux): clear joined flags before channel teardown
b96c571	2026-11-04	A2	feat(dynamic-ux): add named haptics and shared sound service
4de648a	2026-11-04	A2	fix(dynamic-ux): honor vibration preference and serialize sound init
75b1ff8	2026-11-04	A3	feat(dynamic-ux): add per-category notification preferences
72fe744	2026-11-04	A3	fix(dynamic-ux): make notification preferences honest and consistent
f89f540	2026-11-04	A3	fix(dynamic-ux): scope quiet hours to notification feedback
dc5c03c	2026-11-05	A4	feat(dynamic-ux): add local notifications and permission flow
a2c3fc6	2026-11-05	A4	fix(dynamic-ux): harden local notifications and iOS delivery
ed9c0ed	2026-11-05	A4	fix(dynamic-ux): honor sound and vibration per notification
8d8f9e6	2026-11-05	A6	fix(dynamic-ux): unify feedback and remove raw error copy
445c4c9	2026-11-05	A6	fix(dynamic-ux): adopt feedback helper across changed flows
5b97786	2026-11-05	A6	fix(dynamic-ux): block technical error leakage and cover digit validation
da7ff16	2026-11-05	A6	test(dynamic-ux): cover technical error rejection and digit validation
cb7b96a	2026-11-05	B0	feat(qol): add refresh, offline state, badges, and drafts
27c3355	2026-11-05	B0	fix(qol): keep the offline strip out of the app bar
9df1965	2026-11-05	B0	fix(qol): preserve lists and callbacks during refresh
a24048f	2026-11-05	B1	feat(chat): add per-chat mute and manual unread
4b18f8a	2026-11-05	B1	feat(chat): add unread divider, scroll FAB, copy, search, and retry
fdd641b	2026-11-05	B1	fix(chat): own the search controller and wrap menu labels
a25dacd	2026-11-05	B2	feat(qol): add chat actions and activity controls
b8419ca	2026-11-05	B2	fix(qol): surface failures for clear-read and chat state
af0f3c4	2026-11-05	B2	fix(activity): keep the title, targets, and unread swipe
2491fc3	2026-11-05	B3	feat(polish): add resume setup, profile nudge, and empty actions
8110b51	2026-11-05	B3	fix(polish): scope resume state and respect completion
3e11df1	2026-11-05	B3	test(polish): cover completion gate and scoped keys
12afa6e	2026-11-05	B3	fix(polish): keep nudges in layout and unify empty actions
26d38a0	2026-11-05	B3	fix(polish): wrap nudge actions and drop premature dismissal
708316e	2026-11-05	B3	fix(polish): show the profile nudge through the app banner

bb54994	2026-11-05	B3	fix(requests): open sent requests as a backable page
a79ed63	2026-11-05	C0	feat(privacy): add chat status contracts and settings
5c1a4fd	2026-11-05	C0	fix(privacy): mask status data for non-pals and receipts off
4aaaa02	2026-11-05	C0	fix(privacy): make read timestamps rpc-only
4021a43	2026-11-05	C0	fix(privacy): block editing until state loads and reword promise
3fa4a45	2026-11-05	C0	fix(privacy): coalesce in-flight privacy loads
69bec41	2026-11-05	C1	feat(presence): add heartbeat, last-seen, and live online state
c8bb845	2026-11-05	C1	fix(presence): keep heartbeat fresh and gate the channel on loaded privacy
38949d7	2026-11-05	C1	fix(presence): decouple refresh, protect contrast, and scope online
b876970	2026-11-05	C1	fix(presence): stack row status and recover dropped channels
487b85a	2026-11-05	C1	fix(presence): fully tear down failed channels
5829b91	2026-11-05	C1	fix(presence): stop re-entrant channel teardown
27f4784	2026-11-05	C1	fix(presence): guard refresh after sign-out
354ff68	2026-11-05	C2	feat(typing): broadcast throttled typing on the chat channel
8da9b57	2026-11-05	C2	fix(typing): recover typing-only channels and show in empty chats
5bb26dc	2026-11-05	C2	fix(typing): keep the empty match card while a peer types
879e593	2026-11-05	C3	feat(receipts): broadcast and show seen state
3f468e5	2026-11-05	C3	fix(receipts): use the tested payload helper and pause guard
98e3b59	2026-11-05	C3	fix(receipts): trailing read flush and server-authoritative seen
08de0e8	2026-11-05	D	feat(polish): add one-time what's new sheet

Appendix B - File inventory

A = added, M = modified, D = deleted.

State	Path
M	android/app/build.gradle.kts
M	android/app/src/main/AndroidManifest.xml
A	android/app/src/main/res/drawable/ic_stat_palturo.xml
M	ios/Flutter/AppFrameworkInfo.plist
M	ios/Runner.xcodeproj/project.pbxproj
M	ios/Runner/AppDelegate.swift
M	lib/authentication/auth_gate.dart
M	lib/main.dart
A	lib/models/app_notification.dart
A	lib/models/incoming_message_event.dart
M	lib/onboarding/onboarding_complete.dart
M	lib/onboarding/onboarding_flow.dart
M	lib/screen/account/account_settings_page.dart
M	lib/screen/account/change_password_page.dart
M	lib/screen/account/delete_account_page.dart
M	lib/screen/action_dialogs.dart
M	lib/screen/activity_presentation.dart
M	lib/screen/block_list.dart
M	lib/screen/card/request_card.dart
M	lib/screen/chats_page.dart
M	lib/screen/chats_room.dart
M	lib/screen/explore_expand.dart
M	lib/screen/explore_page.dart
M	lib/screen/home_page.dart
M	lib/screen/matches.dart
M	lib/screen/navigation_bar.dart
M	lib/screen/notifications_page.dart
M	lib/screen/profile_edit.dart
M	lib/screen/profile_page.dart
D	lib/screen/request_match_sheet.dart
M	lib/screen/request_page.dart
M	lib/screen/settings.dart
A	lib/screen/settings/chat_privacy_page.dart
A	lib/screen/settings/notification_preferences_page.dart

M	lib/screen/users_dump.dart
M	lib/screen/utils/chat_unread.dart
M	lib/screen/utils/filter_bottom_sheet.dart
D	lib/services/activity_service.dart
A	lib/services/app_badge_service.dart
A	lib/services/app_navigation.dart
A	lib/services/app_navigator.dart
A	lib/services/app_shell_state.dart
A	lib/services/chat_privacy_service.dart
M	lib/services/chat_state_service.dart
A	lib/services/connectivity_service.dart
A	lib/services/draft_service.dart
M	lib/services/haptics.dart
A	lib/services/local_notification_service.dart
M	lib/services/match_api.dart
A	lib/services/notification_presenter.dart
A	lib/services/notification_service.dart
M	lib/services/preferences_service.dart
A	lib/services/sound_service.dart
M	lib/utils/friendly_error.dart
A	lib/utils/presence_label.dart
A	lib/widgets/app_banner.dart
A	lib/widgets/app_feedback.dart
A	lib/widgets/chat_menu_row.dart
A	lib/widgets/chat_mute_sheet.dart
A	lib/widgets/chat_search_sheet.dart
A	lib/widgets/match_banner.dart
A	lib/widgets/offline_banner.dart
A	lib/widgets/resume_setup_card.dart
A	lib/widgets/typing_indicator.dart
A	lib/widgets/whats_new_sheet.dart
M	pubspec.lock
M	pubspec.yaml
A	supabase/migrations/20261105000100_notifications_realtime.sql
A	supabase/migrations/20261105000200_notification_context.sql
A	supabase/migrations/20261105000300_connection_preferences.sql
A	supabase/migrations/20261105000400_clear_read_notifications.sql
A	supabase/migrations/20261105000500_chat_privacy.sql

A	supabase/migrations/20261105000600_privacy_status_masking.sql
A	supabase/migrations/20261105000700_lock_connection_reads.sql
M	test/activity_presentation_test.dart
A	test/activity_qol_test.dart
A	test/app_banner_test.dart
A	test/app_feedback_test.dart
A	test/app_notification_test.dart
A	test/b3_polish_test.dart
A	test/chat_privacy_test.dart
A	test/chat_qol_test.dart
A	test/chat_widgets_test.dart
A	test/draft_service_test.dart
M	test/friendly_error_test.dart
A	test/haptics_test.dart
A	test/incoming_message_event_test.dart
A	test/local_notification_payload_test.dart
A	test/notification_preferences_test.dart
A	test/offline_banner_test.dart
A	test/qol_services_test.dart
D	test/request_match_sheet_test.dart
A	test/sound_service_test.dart
A	test/whats_new_test.dart

Appendix C - Test suite and audit sessions

Test files

File
test/activity_presentation_test.dart
test/activity_qol_test.dart
test/app_banner_test.dart
test/app_feedback_test.dart
test/app_notification_test.dart
test/app_skeleton_test.dart
test/auth_accessibility_test.dart
test/auth_error_test.dart
test/b3_polish_test.dart
test/chat_pending_message_test.dart
test/chat_privacy_test.dart
test/chat_qol_test.dart
test/chat_time_formatter_test.dart
test/chat_unread_test.dart
test/chat_widgets_test.dart
test/draft_service_test.dart
test/friendly_error_test.dart
test/haptics_test.dart
test/home_consistency_test.dart
test/incoming_message_event_test.dart
test/local_notification_payload_test.dart
test/name_change_status_test.dart
test/notification_preferences_test.dart
test/offline_banner_test.dart
test/onboarding_dark_test.dart
test/onboarding_draft_test.dart
test/onboarding_review_test.dart
test/onboarding_semantics_test.dart
test/pals_count_test.dart
test/person_card_skeleton_test.dart
test/person_card_test.dart
test/profile_completion_test.dart
test/profile_edit_test.dart

test/profile_update_link_test.dart
test/qol_services_test.dart
test/request_date_test.dart
test/responsive_shell_test.dart
test/settings_survey_test.dart
test/skills_readonly_test.dart
test/sound_service_test.dart
test/support_page_test.dart
test/system_consistency_test.dart
test/theme_contrast_test.dart
test/validators_test.dart
test/whats_new_test.dart

Audit sessions

Phase	Functionality	Quality	Regression
A0	ses_e592a7f30ffeq8fUwgk2ANDfUM	ses_e592a7f2effeLn8rVWs2rrQPbq	ses_e592a7f2cffe8GfwwpD640Klre
A1	ses_e5901577affeq2wa4Giny1Zc2G	ses_e59015779ffeOb8RHyB6BUeNZe	ses_e59015777ffeEdT9T7s7tkrkxz
A2	ses_e58dab2d7ffeux946Z2eIEhyzD	ses_e58dab2d5ffezFyNm7D75VvxET	ses_e58dab2d3ffeV5XsamgW1U0F5L
A3	ses_e58c5095fffegIgQeovEXrHzdE	ses_e58c5095effeUezhvpt8d7r2sM	ses_e58c5095cffegDv70umgaruv8T
A4	ses_e58adc0d1ffeVWtFpai92ukhG8	ses_e58adc0ceffedzJvP1urIo5eUn	ses_e58adc0caffe3Ux8H0s31HmBs0
A6	ses_e588dbc5affeGDZTXb8Pmwvtnw	ses_e588dbc48ffeXisBr5i8JIysPv	ses_e588dbc45ffesfpI7VGMo097lk
B0	ses_e5843f5e2ffefz7EXGxqIi7bLv	ses_e5836b58bffeKweLZoQIitxgHT	ses_e58300b0bffeTYwA6tCsWyiY2M
B1	ses_e58162853ffeY6eXrwW9epCzOH	ses_e58102699ffe7qOQlEUSgH3ZZD	ses_e58000898ffe0vOEXUSh9pzpQy
B2	ses_e57ef735fffeE9Nt6jlx4U5ATu	ses_e57e2ef40ffeWacaCAK9NOF8US	ses_e57d3680bffeb5fsW9dx9hHZ45
B3	ses_e57c785d2ffe2ZSlK7cqlhTQff	ses_e57bc1a33ffefghRV09XmL0825	ses_e57920c5bffeSS8Rni62EB084P
C0	ses_e577e7e0bffeGsh00O8TG4Aap5	ses_e5771c409ffeX81Tj3KRgUxjeZ	ses_e575d14d5ffeer0oF029QXVh3t
C1	ses_e56cb7616ffekYx0WUloT2M0zV	ses_e56bd7611ffeNnX2Oe0ucXYyKl	ses_e56a791a8ffesn8nk0lS0zWOFT
C2	ses_e5699eeebffef3BcJw4yGqjhig	ses_e56930aa7ffe3BEpK6pz3eqfd4	ses_e568a4196ffeROc75CIa5nBffl
C3	ses_e567e3b64ffe16SQgNbF6cUg0W	ses_e56783fa9ffeX0OJ37fR5RdQMt	ses_e5669814 7ffezN0ys13sCwpNYN

Audit sessions are resumable in the same OpenCode child session if a follow-up is needed.